Lakshya Bajaj

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Education

Purdue University, *West Lafayette, IN* August 2024 – May 2028
B.S. in Robotics Engineering Technology; Concentration: Autonomy and UXVs (Dean's List x 4) GPA: 3.84/4.00
Minors: Electrical Engineering Technology, Business Economics; Certificate: Collaborative Leadership
Relevant Coursework: Applied Computer Vision, Robot Kinematics, Machine Learning & Manufacturing Analysis, Industrial Robot Programming, Design Thinking in Technology

Work Experience

Undergraduate Researcher, Mechanisms and Robotic Systems Lab, *West Lafayette, IN* January 2026 – Present

- Designed schematics and custom PCB in KiCad for a 16×16 piezoresistive tactile array (256 nodes), implementing multiplexed readout and high-speed serial streaming (2 Mbps) for structured force data acquisition
- Developed signal conditioning pipeline including median baseline calibration, per-node thresholding, dynamic range clipping, fixed-scale normalization, and temporal low-pass filtering to improve measurement stability and reduce cross-talk
- Architected multi-threaded real-time visualization system (PyQtGraph + OpenGL) with cubic interpolation (16×16 to 64×64), Gaussian spatial smoothing, and 60 FPS surface rendering
- Extending tactile modeling framework to incorporate shear force estimation alongside normal forces for enhanced 3D force reconstruction in downstream learning-based inference pipelines

Production Engineering Intern, NuTaste Food and Drink Labs, *Gurugram, India* June 2025 – August 2025

- Applied lean manufacturing principles (Kaizen, 5S) and conducted a full Value Stream Mapping exercise to identify and reduce production inefficiencies
- Analyzed cycle time data and performed a cost-benefit analysis on packaging design, raising operational efficiency 4%

Projects

SO-101 Policy Watchdog, *Personal Project* July 2026 – August 2026

- Architected a 6-node ROS 2 system for a SO-101 arm, including a policy-agnostic, vision-based failure detection layer using HSV thresholding
- Trained a custom ACT imitation-learning policy for pick-and-place, recording 50 demonstrations across 5 spatial zones for full workspace coverage
- Designed a supervisor node to arbitrate autonomous and teleoperated control, freezing the arm and handing off to a DS4 controller on detected failure
- Developed DS4 gamepad teleoperation and submitted it as PR #3907 to Hugging Face's LeRobot library, currently open and awaiting maintainer review to be merged

Humanoid Leg Design, *Humanoid Robot Club* August 2024 – May 2025

- Contributed to the award-winning "chicken leg" foot architecture, earning the AMD Most Innovative Design of the Year award for mechanical–electrical integration and stability-driven design innovation
- Researched and designed electrical subsystems for humanoid legs, integrating motor drivers, IMUs, and microcontrollers for closed-loop control, real-time state estimation, and hardware validation during dynamic locomotion testing

Leadership Positions

Team Leader International, Purdue Orientation Programs, *West Lafayette, IN* March 2025 – September 2025

- Mentored and guided 35 first-year students during Purdue's nationally recognized freshman orientation program
- Facilitated icebreakers, discussions, and team-building activities to ease cultural adjustment and build community
- Oversaw integration of international and domestic students, fostering cross-cultural connections and peer support

Skills

Robotics, Control & Automation: Industrial Robot Programming (FANUC KAREL, Yamaha Robot Language), ROS2, Kinematics, Motor Control, OpenCV
Electronics & Hardware: Circuit Design, PCB Design and Fabrication (KiCad), Signal Conditioning, Power Systems
Embedded Systems & Sensing: Arduino, Raspberry Pi, Microcontrollers(ATMega 2560), Microchip Studio
Programming: Python, C, MATLAB
Mechanical Design & Prototyping: Fusion 360, Siemens NX & Teamcenter, 3D Printing, Laser Cutting